# How to Close the Digital Divide in the U.S.

**by Bhaskar Chakravorti**

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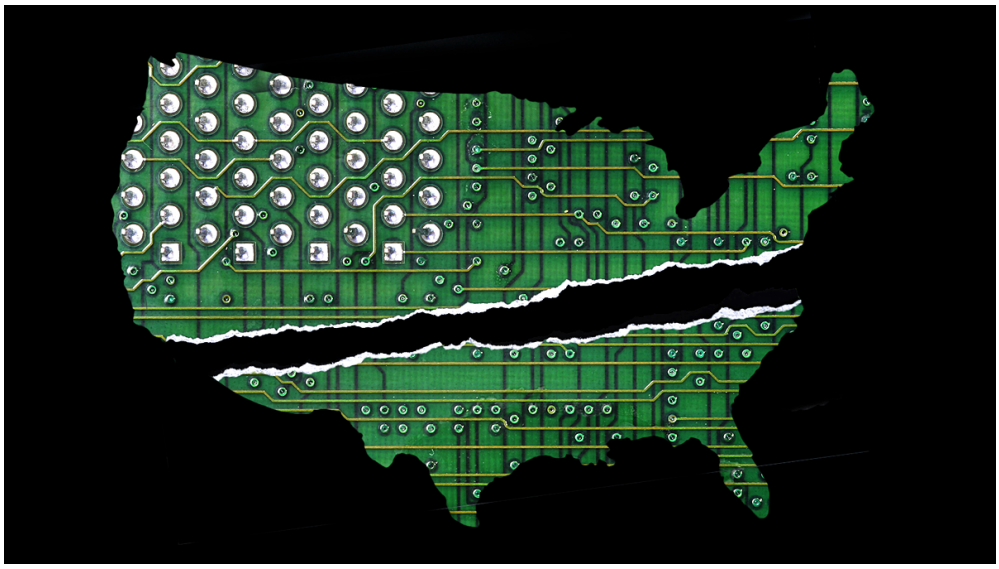

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**Finally, America appears to be taking bold** action on fixing its fraying infrastructure. President Joe Biden's proposed American Jobs Plan — despite being negotiated down in a bipartisan deal — is a significant step in addressing one the of the country's most pressing, deeply rooted, and often overlooked problems: The plan contains a $65 billion budget spread over eight years to close the gaps in the digital infrastructure.

It's a major investment. Unfortunately, it still falls well short of what's needed to actually solve the problem. According to our analyses, the

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budget should have been *two and half times* as large as the original $100 billion. Let me explain why and what can be done to move forward.

It's long been acknowledged that even as the digital industry exploded out of this country, America lived with a "digital divide." While this is loosely understood as the gap between those who have access to reliable internet service and those who don't, the true nature and extent of the divide is often under-appreciated. Internet infrastructure is, of course, an essential element of the divide, but infrastructure alone does not necessarily translate into adoption and beneficial use. Local and national institutions, affordability and access, and the digital proficiency of users, all play significant roles — and there are wide variations across the United States along each of these.

As part of our research initiative, *Imagining a Digital Economy for All (IDEA) 2030* (established in collaboration with the Mastercard Center for Inclusive Growth), we disaggregated the digital divide into four distinct components and scored the 50 states along each of these:

- *Infrastructure:* Internet speeds; terrestrial broadband coverage; smartphone usage.
- *Inclusivity:* Affordability of broadband; equity of broadband access across income groups; actual usage of the internet at broadband speeds.
- *Institutions:* Political prioritization of broadband strategy; best practices of government use of technology for public services; restrictions on alternative local broadband solutions, such as municipal networks.
- *Digital Proficiency:* How well people can navigate the digital world, which is shaped by demographic profile, education levels, political tolerance, degree of skepticism about news sourced from social media.

The state of the digital divide across all 50 states can be visualized in the exhibit below.

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## Charting the U.S. Digital Divide: How the 50 States Stack Up

Researchers scored the 50 states across four key measures: infrastructure, institutions, inclusivity, and digital proficiency. The sum effect of these measures paints a picture of each state's particular set of challenges when it comes to closing the digital divide and forms the basis for each state's overall score. For example, while West Virginia has offered incentives for remote workers — and its institutions score reasonably well — its poor infrastructure, inclusivity, and digital proficiency scores indicate that deep investments across those measures will be required to close its digital gap.

Source: Digital Planet, The Fletcher School, Tufts University

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## The Digital Divide Reinforces Divides in Key Services

The inability to use the internet pushes access to many essential services out of reach. Often, this compounds other inequities and historical injustices.

Consider telehealth, an essential lifeline post-pandemic — particularly with a surge in mental health consultations. Even as many states expanded telehealth policies to accommodate this surge, our research shows that the digital divide hampered access in New Mexico, Montana, Vermont, and Iowa. Other states, including West Virginia, Alabama, Oklahoma, Indiana, Missouri, Tennessee, and South Carolina, were slow in expanding their policies and are poor in terms of digital access as well.

A parallel story played out with schools. As more than 55 million students moved to online learning during the pandemic, one in five teens, ages 13 to 17, reported being unable to do their homework "often" or "sometimes" because of unreliable Internet access. Twelve million children were without internet access altogether. The challenges varied by location. To consider an extreme example, 70% of children in the Kansas City school district did not have internet access at home, a problem made worse by the fact that Missouri only spends about $10,600 per pupil, which puts it near the bottom third compared to other states.

The digital divide also reinforces racial inequity. Nearly half of Americans without at-home internet were in Black and Hispanic households. With a 14-point gap in broadband access between white and Black households with school-going children, and a 12-point gap between white and Hispanic households, we find that up to 40% of disconnected K-12 students from Black, Latino, and indigenous communities struggle with insufficient digital literacy, language obstacles, and other disincentives to use the internet and find ways to gain better access.

The digital divisions are likely to keep these historic disadvantages in place in the future. Seventy percent of Black and 60% of Hispanic respondents

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report being underprepared with digital skills, affecting their employability. While a third of all white workers in 2018 were in jobs they could do from home, less than 20% percent of Black workers and only 16% percent of Hispanic workers were in jobs that could be done remotely. Without additional intervention to close this divide, a majority of Black and Hispanic workers could be locked out of 86% of jobs by 2045.

The divide has economic costs overall. Access to reliable internet is also a strong predictor of economic opportunity. According to a Deloitte study, a 10% increase in broadband access in 2014 would have resulted in more than 875,000 additional U.S. jobs and $186 billion more in economic output in 2019. The shift to remote work has been an opportunity to spread talent and economic benefits across the country — 14 to 23 million Americans say they intend to relocate to a different city or region, according to a study by Upwork, and many already have. But while several cities and states have been offering incentives for inbound moves, most of the regions eager for inbound moves also have some of the largest digital gaps. Consider West Virginia, where the governor is offering $12,000 to remote workers. But according to our analyses, West Virginia ranks 50 out of 50 in infrastructure and digital literacy, 46 in inclusivity and 19 in institutions. Sixty-two percent of urban West Virginia does not use the internet at broadband speeds.

## The Digital Divide is Deeper Than We Think — But We Can Still Fix It

Biden's original infrastructure budget proposed $100 billion for digital infrastructure. While large, it is actually not large enough: It mirrored a parallel proposal in Congress, which, in turn, drew upon a 2017 FCC estimate that it would cost $80 billion to expand broadband access to every household. These budgets use an incorrect FCC mapping of access to the country's digital infrastructure, which pegs the broadband disconnected at "fewer than 14.5 million," which even the current acting FCC chair, Jessica Rosenworcel, acknowledges is an undercount. A more reliable "manual" check by the research group, BroadbandNow, estimated

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42 million Americans were without broadband; factoring in other challenges in getting people to actually use the service, that number of people not using broadband is arguably much higher.

Using the FCC's cost structures on these revised figures, our research team analyses the budget need to be at least $240 billion — $175 billion more than the $65 billion allocated under the terms of the current bipartisan deal. The number could be even higher, as the Biden American Jobs Plan fact sheet calls for "future-proof" broadband and the need to upgrade the nation's broadband standards to enable high bandwidth-using applications such as streaming video and Zoom conferencing that have proved to be essential through the pandemic. While the Biden team insists it has creative ways to stretch the tighter budget, it still needs new revenue sources and opportunities to save on costs.

Fixing the digital divide ought to be a priority as it sits at the center of numerous other societal problems, ranging from racial inequities to unevenness of access to essential needs, including health care and education. The execution will need to be locally appropriate and must go beyond filling in physical infrastructure.

I have several recommendations for action. They will require leadership and collaboration across government and business at the federal and local levels.

1. **Use a "Romer" tax to cover the budget shortfall.** The current political environment demands that this initiative be paid for without deficit spending. I would recommend turning to the industry that stands to benefit the most from connectivity: Big Tech. The Nobel laureate, Paul Romer, has recently suggested taxing revenues from targeted digital ads, which, in my opinion, would be an ideal revenue source. Digital ads offer large revenue pools: In 2020, social media advertising revenues were $41.5 billion, while digital video advertising revenue were $26.2 billion and advertisers were expected to

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spend $59.22 billion on search ads. A tax rate of, say, 19% could help close the $175 billion budget gap over eight years. The tech tax revenues could be collected in a new Universal Broadband Fund, modeled on the Universal Service Fund, by which long-distance telecoms were assessed to subsidize telephone service to high-cost areas. FCC chair Rosenworcel has said she finds such a proposal "intriguing," but that "it's clear that this would require action from Congress." Maryland has already adopted the idea and other states are considering it.

2. **Coordinate locally appropriate solutions.** The digital divide is an aggregate of many divides, with local barriers to be overcome. We should prioritize efforts in states where less than a third of the population has basic broadband access: Arkansas, Kentucky, Mississippi, New Mexico, Alabama, and West Virginia. We need to enable new entrants by removing bureaucratic obstacles, such as bans on municipal networks, which currently exist in 18 states. All states should have a broadband strategy. In addition, a reconsideration of the FCC's 2017 decision to roll back net neutrality — the idea that internet service providers should not be allowed to favor or throttle service for particular forms of traffic — can help restrain broadband providers from charging more for certain services or content. A federally orchestrated approach will help ensure consistency, inclusion, and a prioritization of resources and regulatory reforms.
3. **Recruit Big Tech and internet service providers (ISPs) to help close gaps.** As the government has leverage over the tech giants that are in the regulatory crosshairs — Facebook, Alphabet, Amazon, and Apple — these companies ought to be encouraged to offer favorable deals on the internet access infrastructure they own. For example, Facebook has high-capacity fiberoptic networks that can be used to provide broadband access, along with its Terragraph technology that can be used to deliver fiber-like speeds in urban settings. In parallel, major ISPs should also be made to fulfill their outstanding commitments to provide internet access. CenturyLink and Frontier Communications, for example, have taxpayer-subsidized broadband

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connectivity projects that need to be finished. These "low hanging fruits" should be a key part of the administration's plan.

4. **Identify gap areas and invite public-private solutions**. Given the numerous restrictions on purely public or municipal solutions, the federal government should be open to local public-private partnerships prioritizing the most vulnerable areas to accelerate the process. It can organize a bidding process to solicit solutions and get state governments involved in the process, then set targets for each state and tie subsidies, grants, and additional incentives to hit them. There is successful precedent for similar projects: Google and the state of California collaborated to connect 100,000 rural households, Microsoft and other companies worked with NGOs on a digital connectivity pilot in East Cleveland, and Facebook's fiber network helped connect multiple educational institutions in North Carolina.
5. **Update and expand existing affordability programs**. The FCC's E-Rate program, originally designed to provide discounts to schools and libraries for telecoms access, should be expanded to include households with schoolchildren in broadband deserts. Then there's the federal Lifeline program, designed to provide subsidies for telephone and internet services to eligible low-income households. However, the service quality was so poor that only 25% of those eligible took advantage of it. This situation can be improved by providing further funding for Lifeline and offering a wider set of better connectivity plans to users.
6. **Build in future-proofing**. The standards for broadband need to be raised in order to accommodate the increased demands on internet ecosystems down the road. A bipartisan group of four U.S. senators have even sent a letter to the FCC, the U.S. Commerce Department, and the Department of Agriculture, urging raising the federal broadband standard to 100 Mbps in keeping with these growing needs. Resistance from cable and telecom companies about such a change in standards needs to be anticipated and managed. (These companies worry that they would be excluded from government grants and subsidies and that competitors backed by public subsidies would have an unfair

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advantage.) Also, future-proofing should integrate a federal "Dig Once" mandate that would require the simultaneous construction of broadband infrastructure alongside other unrelated construction projects, wherever feasible. Such a policy could save considerably on construction costs and allow simultaneous construction of broadband infrastructure alongside other infrastructure projects.

7. **Invest in digital literacy**. Only 40% of American adults can answer basic questions on topics including phishing, privacy, and cookies. Digital literacy is key to improving adoption, combating misinformation and scams, and limiting the risks of cyber-attacks. Public and private-sector initiatives can build on the foundations set by the Accessible and Affordable Internet for All Act that puts aside $60 million in grants to states seeking to bolster digital literacy programs. Lifelong digital proficiency programs — beginning at a young age in the broader education system and continuing through later adulthood on the job — can offer Americans a source of resilience against these threats and make them better consumers of the digital ecosystems. More resources for "digital inclusion funds" and "digital navigators" that already help low-income households and older adults in the use of technology in many cities ought to be scaled-up nationwide

Bridging of the digital divide is both complex and daunting, but there are reasons for hope. Many institutions — both public and private — stand to gain from it being addressed. The pandemic experience is a reminder of the very real costs of the digital divide that have added up over the years: Schoolchildren without internet access fell behind. Residents of internet deserts missed vaccine appointments snapped up by internet-enabled non-residents. Disadvantaged minority populations felt the pinch of inadequate high-speed internet access as jobs and job searches went remote. We now have a rare bipartisan agreement on the urgency of solving the problem. Let's act on it.

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*digital divide in the United States, please visit Turning America's Digital Divide into Digital Dividends.*

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**Bhaskar Chakravorti** is the Dean of Global Business at The Fletcher School at Tufts University and founding Executive Director of Fletcher's Institute for Business in the Global Context. He is the author of *The Slow Pace of Fast Change*.

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